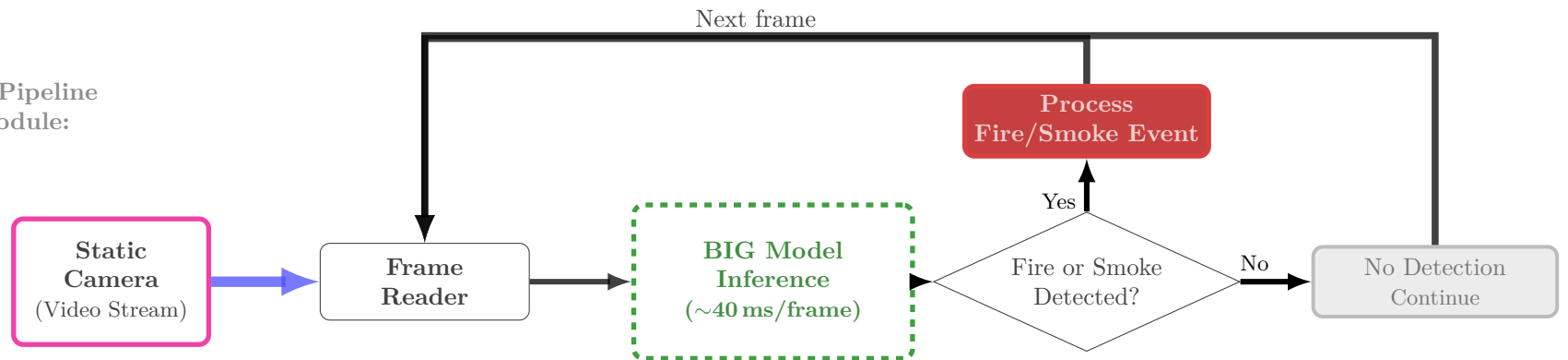
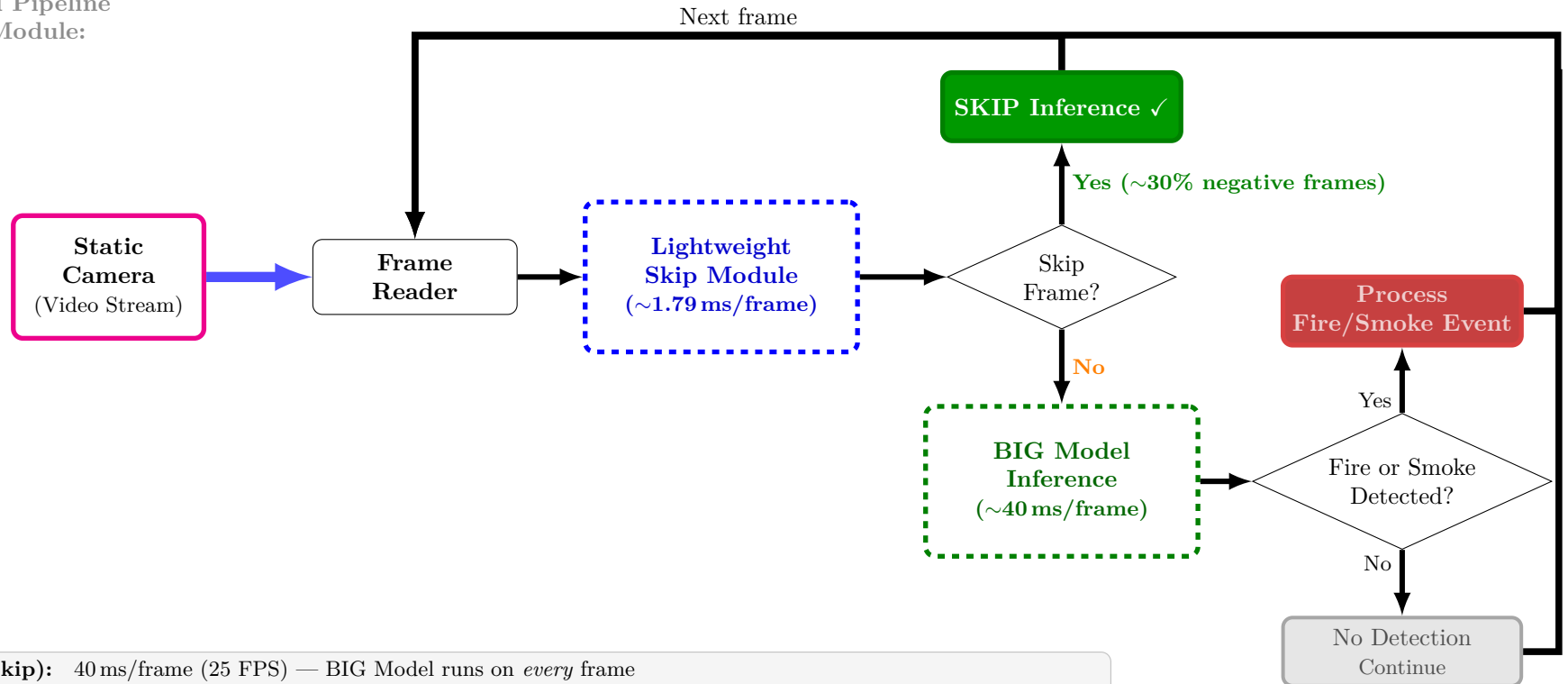


(a) Baseline Pipeline  
- No Skip Module:



*~ 40 ms/frame – ~25 FPS (runs on every frame)*

(b) Proposed Pipeline  
-With Skip Module:



**Baseline (no skip):** 40 ms/frame (25 FPS) — BIG Model runs on *every* frame

**Proposed (with skip):**  $= (0.27 \times 1.79) + (0.73 \times 41.79) \approx 31.0 \text{ ms/frame } (\approx 32 \text{ FPS})$       **Speedup:**  $\approx 1.3\times$  (+29% FPS)

*Assumes 90% of all frames are negative, the skip rate of negative frames is 30%.*